

AI-Based Stock News Sentiment Tracker

Problem:

Investors often react quickly to financial news, but it is difficult to read and understand hundreds of articles every day. This project uses AI to automatically collect news articles about selected companies and summarize the overall sentiment so investors can quickly understand whether coverage is positive, negative, or neutral.

Crawler Task:

The crawler collects financial news from websites like Yahoo Finance, Reuters, and CNBC. It searches for articles related to five stocks: MSFT, GOOGL, AAPL, TSLA, and META. For each article, the system extracts the headline, article text, publication date, and link. This allows the system to gather relevant information automatically each day.

AI Task:

After collecting the articles, an AI model performs sentiment analysis to classify each article as positive, negative, or neutral. The system also identifies keywords that help explain the sentiment, such as “earnings,” “growth,” “regulation,” or “AI.”

A Generative AI tool is then used to create a short insight summarizing the daily news sentiment for each stock.

Output:

The system produces a daily sentiment summary for each stock, including:

- Number of articles collected
- Overall sentiment (positive, neutral, or negative)
- Keywords driving sentiment
- A short AI-generated insight

Example insight:

“TSLA sentiment turned negative after earnings announcement.”

Selected Stocks:

- Microsoft (MSFT)
- Alphabet (GOOGL)
- Apple (AAPL)

- Tesla (TSLA)
- Meta (META)